CVO1 – AGREEMENT ACROSS COVERS AND COMMON TUNING

Abstract. We outline the supplied comparison of QED3 local constructions across odd covers and the selection of a common bare mass and vacuum pair. A single top primitive system is projected to its quotients. Exact agreement on flat lifts then transports the pair selected on the top torus to the prescribed smaller tori. The opening theorem states the actual-map hypotheses, stopping thresholds and bounds; linked proof cards identify the arguments and their sources.

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1. Introduction and target statement

CVO1 treats S16–S18 and T01 of QEO1 within its renormalization stage. The result concerns the prescribed odd reference tori.
Current proof manuscript: CVM1, Agreement across covers and common tuning for lattice QED in three dimensions.
Source editions. The 263-page master and its TeX source are the preserved comprehensive edition of QEM1. Its page, section and line locators below refer to that edition. Current companion manuscripts are cited by their codes and result names.

Original inputs: S1, curvature limits and the fourth cumulant; S2, chamber fermions and reflection positivity; S3, chamber renormalization and the fourth cumulant.

T01. One pair on the prescribed odd tori

Model-specific. Overview Target.

Physical setting. Fix an odd integer $L \geq 13$, an integral power M of L with $M \geq L^2$, an integer $r \geq 2$, $0 < \varepsilon \leq 1/200$, $c_{\mathrm{fl}} \geq 1$, and $0 < \bar{m} \leq 1$. For an integer $N \geq 0$, set

$$h = L^{\lfloor \log_L(1/\bar{m}) \rfloor}, \qquad \eta = L^{-N}, \quad K = N + \log_L h, \quad a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \mu_j = \bar{m} a_j.$$

Here $0 \leq j \leq K$, and $q = e\sqrt{\eta}$ denotes the charge at level zero. We write $\bar{e}_0, \bar{e}_1$ for the fixed charge ceilings, to distinguish them from the running charges e_j. Put $v_j = -\log e_j$, and let r_j be the least power L^i, $i \in \mathbb{Z}_{\geq 0}$, with $r_j \geq v_j^r$. The physical sides are

$$R_{\mathrm{top}} = hMr_K, \quad R^* = R_{\mathrm{top}}/L, \qquad \mathcal{T}(e) = \{hL^T : T \in \mathbb{N} \cup \{0\},\ hL^T \leq R_{\mathrm{top}}\}.$$

They are independent of N. For $R \in \mathcal{T}(e)$, define $\ell_j = R/a_j$ and $n_j = \log_L(\ell_j/(Mr_j))$. The stopping level $k_*(R)$ is the first j with $n_j = 0$, when it exists.

Numerical hypotheses. Assume

$$0 < e\sqrt{h} \le \bar{e}_1 \le \min(\bar{e}_0, 1/3), \qquad \bar{e}_0^{\varepsilon} \le 1/4,$$
$$c_{\mathfrak{fl}} L^{-3/8+6/200} \le 1/16, \qquad L^{-5/8} \le 1/4,$$

and, for every $v \ge -\log \bar{e}_1$,

$$v \ge 8, \qquad \left(1 + \frac{3\log L}{2v}\right)^r < L, \qquad v^r > 31L + 16,$$
$$(c_{\mathfrak{fl}} + 2)e^{-\varepsilon v} \le \frac{1 - L^{-\varepsilon/2}}{16}. \tag{1}$$

Actual maps and imported hypotheses. Use the real chamber activity spaces and their independent connection slots recalled in D. Write $\mathscr{E}_j$ for an activity with external slot C, and $E_j = \mathscr{E}_j|_{C=0}$. At every used scale assume the exact primitive, determinant, extraction, reconstruction, contour-domain and convergent-logarithm contracts listed in I. These are the actual maps of QEO1/S15, with the domains and numerical prerequisites in S05, S08, S09, S10, S11, and S13–S14. On the top torus choose the photon and fermion primitive families of S16; on its smaller odd quotients use their projections. The incoming and outgoing sides must lie in the admitted class $\ell/(LM)$ odd and at least 13. At a compared step require $\ell_{\mathrm{small}}/L > (31L + 16)M$, as in S18.

Here are the quantitative consequences of these imports used in tuning. Set $w_j^m = e_j^{3/4-8\varepsilon}$ and $w_j^E = e_j^{1/4-7\varepsilon}$. The no-slot data norm is $\max\{|m|/w_j^m, \|E\|_j/w_j^E\}$; with an external slot, replace $\|E\|_j$ by $\|\mathscr{E}\|_j^x$. Radius-four data balls mean the corresponding closed balls in the complexified spaces. Unit balls below are closed balls in their real subspaces. At zero the activity coordinate is fixed to zero. The maps preserve reality and obey

$$m_{j+1} = L(m_j + m(E_j)), \qquad \epsilon_{j+1}^v = L^3(\epsilon_j^v + \epsilon_v(E_j) + \delta\epsilon_j^0), \tag{2}$$
$$E_{j+1} = \Phi_j(m_j, E_j), \qquad \Phi_j(m, E) = \mathcal{L}(RE)^{\mathrm{nat}} + \mathcal{L}E_j^{\mathrm{det}} + E_j^*(m, E). \tag{3}$$

Restriction to $C = 0$ commutes with the full external-slot recursion and does not increase its norm. Scalar extraction is linear and real,

$$|m(E)| \le c_{\mathfrak{fl}} e_j^{1/2} \|E\|_j, \qquad |\epsilon_v(E)| \le c_{\mathfrak{fl}} \|E\|_j, \qquad 0 \le \delta\epsilon_j^0 \le e_j^7.$$

On the unit ball the normalized size and difference estimates are

$$L^{-1} w_{j+1}^m / w_j^m \le 1/4,$$
$$\|\Phi_j(m, E)\|_{j+1}/w_{j+1}^E \le \tfrac{1}{16}\|E\|_j/w_j^E + 2e_{j+1}^{\varepsilon},$$
$$|m(E) - m(\widetilde{E})|/w_j^m \le c_{\mathfrak{fl}} e_j^{\varepsilon} d_E,$$
$$\|\Phi_j(m, E) - \Phi_j(\widetilde{m}, \widetilde{E})\|_{j+1}/w_{j+1}^E \le \tfrac{1}{16} d_E + e_{j+1}^{\varepsilon}(d_m + d_E), \tag{4}$$

where $d_m = |m - \widetilde{m}|/w_j^m$ and $d_E = \|E - \widetilde{E}\|_j/w_j^E$. The same size estimate holds for the external-slot norm. The fixed-point argument uses the no-slot difference estimate. Initially $E_0 = \mathscr{E}_0 = \mathfrak{P}_0 = 0$, where $\mathfrak{P}_j$ is the scalar external family in S18. At zero the activity coordinate is fixed to zero; the source's first-step chamber definitions in I apply.

Target theorem. For each $R \in \mathcal{T}(e)$ there is a finite threshold $N_{\min}(R, e)$ such that the first stop exists for $N \ge N_{\min}(R, e)$. Its physical quantities are then independent of N. The thresholds are nonincreasing and the stopping levels nondecreasing with R; their precise formulas are in T01.a.

For $N \ge N_{\min}(R_{\mathrm{top}}, e)$, there is exactly one real pair $P_N(e) = (m_{0,N}(e), \epsilon_{0,N}^v(e))$ whose top-torus trajectory is controlled and has $m_{k_{\mathrm{top}}} = \epsilon_{k_{\mathrm{top}}}^v = 0$. Controlled means that each preceding step is defined in its radius-four data ball and $|m_j| \le w_j^m$ through the stop. For every $R \in \mathcal{T}(e)$ with $N \ge N_{\min}(R, e)$, the same pair is controlled on R. Up to its stop, the scalar mass and vacuum

sequences equal the top sequences, while the activities and external scalar families agree on admissible flat lifts under S17's algebra identification. On each such torus,

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon}, \qquad |\epsilon_j^v| \le 5c_{\mathfrak{fl}} e_j^{1/4-6\varepsilon}. \tag{5}$$

The smaller terminal couplings need not vanish.

Within the fixed construction, the pair is determined by $(N, e, \bar{m})$ and is independent of the smaller side, twist, external connection and probe. Define only $m_{0,N}(e) = 0$ below the top definition threshold. For every $N \ge 0$,

$$m_{0,N}(e) \in \mathbb{R}, \qquad |m_{0,N}(e)| \le \tfrac{1}{8} q^{3/4-8\varepsilon} < \tfrac{1}{8}.$$

The total bare mass $\boldsymbol{\mu}_N = \bar{m} L^{-N} + m_{0,N}(e)$ has no asserted sign. On R^*, for $N \ge N_{\mathrm{ref}}(e) := N_{\min}(R^*, e)$,

$$a_* \in \{h/L, h/L^2\}, \qquad \ell_* = M r_*, \qquad r_* > 31L + 16, \qquad L^{-3} < \mu_* \le L^{-1}.$$

Proof organization. The expanding mass recursion is solved backwards and the contracting activity recursion forwards on one finite sequence. T01.b selects the top pair and recovers its vacuum by a scalar sum. The cover argument has a different task: S16 constructs related primitive systems, S17 identifies their supported rows, and S18 compares the actual local operations. This exact comparison, together with continuation on each smaller torus, transfers the chosen pair. Equality on flat polymers does not bound an entire smaller-volume activity; its own continuation estimate supplies that bound.

Transfer to the smaller tori. Choose the pair on R_{top} once. Start every eligible smaller torus with this pair and with zero activity and scalar external family. Induct simultaneously on equality of the scalar couplings, existence of the smaller trajectory, and local agreement of its two families. At a given level, the mass equals the already controlled top mass, so continuation constructs the next smaller step with its global norm bound. The side calculation in T01.a permits S18. That comparison gives equal extracted scalars and box cost, and local agreement of both outgoing families. This closes the induction. It stops at the smaller terminal level, where the couplings are the corresponding top values and need not vanish.

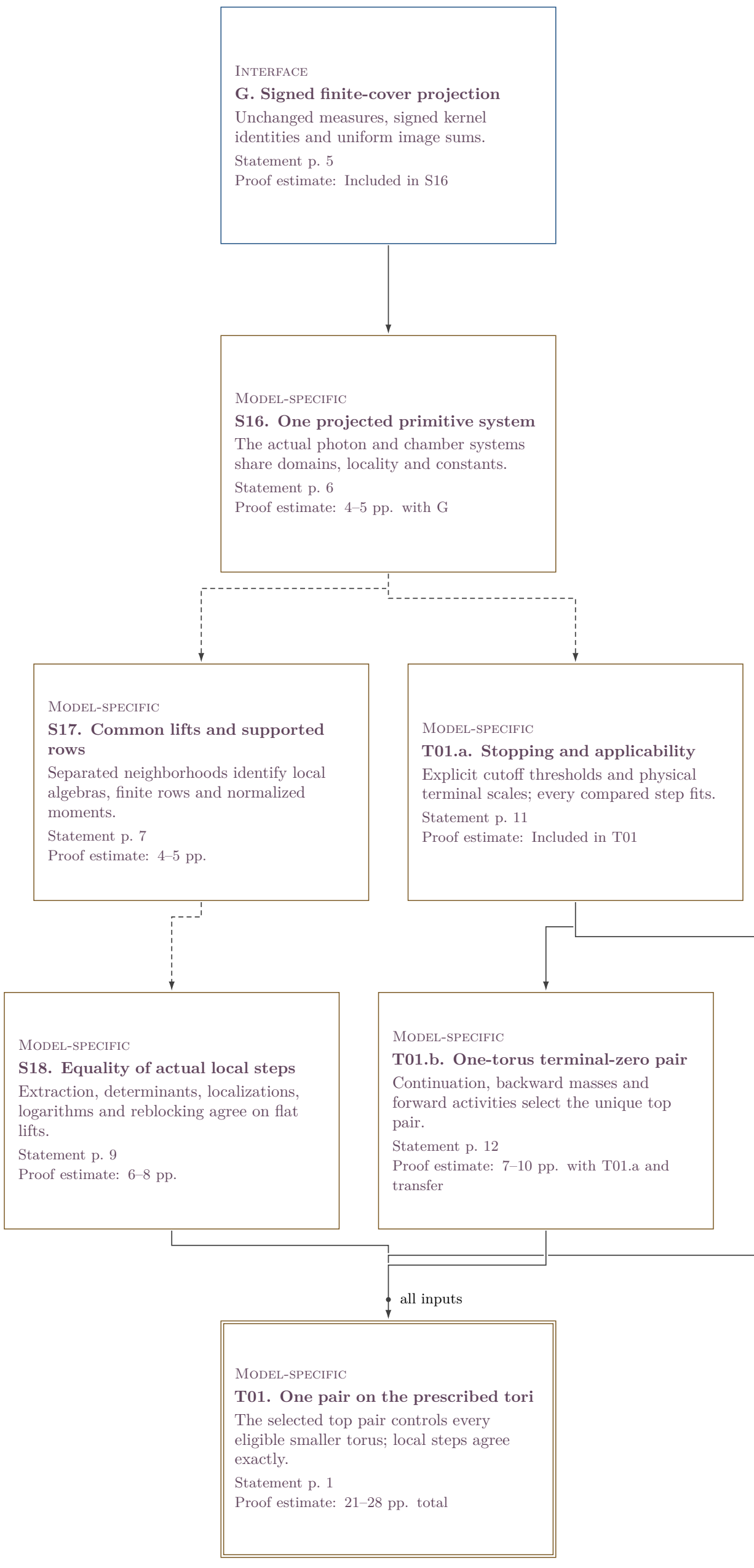
The common pair is therefore selected before a smaller side, twist, external connection or probe is inserted. Its dependence on N, e and the fixed construction remains. The source's zero extension below the top threshold applies only to the mass; it gives the all-cutoff bare-mass statement above. The transfer is 263-page master §16.4, Theorem 16.4, and S3 Proposition 28.3.

Source and scope. 263-page master §§15–16, especially Theorems 15.5, 16.3 and 16.4; original S3 Propositions 27.6 and 28.1–28.3 and their supplied proofs. The current proof manuscript is CVM1. The original proof allocation was 14–18 pages for the cover group and 7–10 for stopping and tuning, 21–28 pages in total. These ranges count shared setup once and exclude the imported block and nonlinear step proofs. The target concerns the prescribed odd reference family; it supplies parameters for later construction volumes without claiming a controlled renormalization trajectory on arbitrary even tori.

CVO1 – Agreement across covers and common tuning

Definitions/setup Model-specific Interface (general)

Standing inputs: T01's physical and numerical hypotheses; the actual primitive and nonlinear estimates in I; the activity domains in D. The quantitative step bounds enter both S18 and T01.b.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.
Click a box for its statement. Return links are navigation, not proof dependencies.

T01.a and T01.b are subordinate proof cards. Final transfer uses all three branches: stopping checks, the selected top trajectory and exact comparison. Budgets are the original source-based planning estimates and count shared setup once. Current CVM1 is linked from the introduction.

2. G. Signed finite-cover projection

Interface. Overview Target.

Objects and hypotheses. Let $\ell_2 = J_c\ell_1$, J_c odd, with both $\ell_i/(LM)$ odd and at least 13, at common mesh $\eta = L^{-k}$. The column character on the covering index set is $\chi(w) = 1$ for photon fields, coordinates and parameters, and $\chi(w) = (-1)^{w_1+w_2+w_3}$ for spinors and pair fields, where $w \in \mathbb{Z}^3$. Pullback p^* identifies small-torus fields with the corresponding character subspace of the large torus. Matching column measures are unchanged: 1 on unit indices, η^3 on fine photon indices, and $\eta^3/48$ on chamber indices. A top parameter family is translation covariant under $\ell_1\mathbb{Z}^3$ on pulled-back data. For such a family put $\widetilde{O}(s, A) = O_2(p^*s, p^*A)$. For a polymer family assume translation covariance with the polymer translated together with its row, column and connection data, and put $\widetilde{O}(X, A) = \sum_{pX'=X} O_2(X', p^*A)$. Assume its row/column support diameter is at most L, $\gamma_w M \geq 1$, $\kappa_w \geq 0$, and the rooted polymer counting bound with constants κ_0, K_0.

Claim. $\|p^*f\|_\infty = \|f\|_\infty$ and pairings multiply by J_c^3. Choose one integer representative of each class in $(\mathbb{Z}/J_c\mathbb{Z})^3$, forming a set $\mathcal{R}$. The projected operator has the exact kernel

$$O_1(t, t') = \sum_{w\in\mathcal{R}} \chi(w)\widetilde{O}(t, t' + \ell_1 w).$$

Use the character of the operator's column family. The combined summand, not the antiperiodic sign alone, is independent of the representative. Indeed replacing w by $w + J_c v$ multiplies both its sign and its large-torus kernel by $\chi(v)$, so their product is unchanged. Intertwining passes to transposes and to inverses when defined. Projection of a represented net gives $d_M(pX') \leq d_M(X')$. The uniform image estimates are

$$\sum_w e^{-\gamma_w d_2(t,t'+\ell_1 w)} \leq 64e^{-\gamma_w d_1(t,t')},$$

$$\sum_{\substack{pX'=X\\ X'\supset\square_t}} e^{-(\kappa_w+\kappa_0)d_M(X')} \leq K_0 e^{-\kappa_w d_M(X)}.$$

Geometric convention. Here d_i is periodic sup distance between the supports of the indicated indices. The polymer length $d_M(X)$ is the infimum of the total represented segment length, in units of M, of connected nets meeting every cube of X. The counting input is $\sum_{X'\supset\square} e^{-\kappa_0 d_M(X')} \leq K_0$. One may use $\kappa_0 = 54\log 72$, $K_0 = e^{\kappa_0}/36$. Projection is applied to a segment representation before taking this infimum.

Proof outline. Decompose a large fundamental set into J_c^3 small translates. The column weights give the pairing identity and signed kernel formula. Translation covariance preserves the relevant character subspace; the pairing identity then proves transpose intertwining. Invertible intertwining gives inverse intertwining. Project represented net segments to obtain the net-length inequality. For the kernel sum choose nearest period representatives. There are 27 near representatives and at most $26j^2$ in the sup-radius shell $j \geq 2$; the latter have extra exponential decay at least $6(j-1)$. The resulting bracket is below $27.3 < 64$. In the polymer sum use the projected net bound for the κ_w factor and spend the remaining κ_0 on rooted counting. No limiting volume is involved.

Source. 263-page master §§15.1–15.2, Proposition 15.1 and (15.1)–(15.5). S3 Propositions 26.1 and 26.3(a),(d),(e), with their proofs in Chapter 26. The general calculation is separated here from its model verification in S16. Its proof allocation is included in S16's 4–5 pages.

3. S16. One projected photon and fermion system

Model-specific. Overview Target.

Hypotheses. Use the covers, measures and counting constants of G. Assume the exact photon and chamber primitive constructions in I, including their inverse, walk, domain and charge conditions. In particular $M \geq L^2$, $M > 4L$, $M \geq M^P$, and $\gamma_w M \geq 1$; M^P is the photon threshold in the imported construction. The top walk is constructed with decay $\kappa_w + \kappa_0$, not merely κ_w. All smaller systems are projections of this single top system.

Model verification and output. The exact photon minimizer, coarse form, axial coordinate map, fluctuation form and its resolvent/root, fluctuation injection and constrained block minimizer all intertwine with pullback. The exact chamber/fine/coarse fermion operators and their inverses do likewise, on the pulled-back local-real-gauge domain. A top weakened system with kernel constant $C_w^{(1)}$, rates $\gamma_w, \kappa_w + \kappa_0$, analytic parameter polydiscs, full-strength identities, active-component restriction, local read-dependence, H_k-row vanishing and the required covariances projects to the same properties with rates γ_w, κ_w and common constant $C_w = 64 K_0 C_w^{(1)}$. At zero preserve the normalized chamber measure and $(1+y)^{-1}$ resolvent factor.

A concrete top choice is furnished by the path-multiplied photon kernels and weighted fermion walks, with

$$\alpha_0 = 1/54, \qquad \gamma_w = \min(\gamma_F, \gamma_P/2),$$
$$C_P = 116 C_P^{\mathrm{ph}} M^{402\alpha_0}, \qquad C_w^{(1)} = \max(C_F, C_P, 4\exp(2\gamma_w) M^{27\alpha_0}).$$

Here the top fermion walks reserve the additional κ_0 decay; their M and charge conditions and the photon path-count condition are imposed before choosing the charge ceiling. The constants are independent of side, level, admitted mass, charge and real harmonic background.

Proof outline. For the exact photon minimizers, pull back the Euler problem and use uniqueness. Apply the positive resolvent integral to the fluctuation root. For the fermion operators, compare the chamber stencil, J, J^*, the six-path averages, penalties and finite block resolvent factors term by term. A five-cube neighborhood has a nonwrapping lift of side $11M < \ell_1$; the pulled-back connection therefore remains in its declared local-real-gauge domain. Only then apply inverse intertwining.

Apply G's two summation estimates to the top weakened kernels. Check analytic parameter polydiscs, connection analyticity and resolvent-parameter Borel measurability together with the kernel bounds. For a supported row first restrict to its attached active component. The component and the missing-cube vanishing condition project to the corresponding conditions below. For a polymer kernel, both indices belong to the same supported preimage, so column support also survives. Gauge, lattice, charge and reality covariance commute with projection.

At zero the common parameter products retain all 48 chamber labels, and the column weight stays 1/48. The local resolvent retains $(1+y)^{-1}$; projection introduces no additional factor 48. The path-multiplied photon construction and weighted fermion walks provide the top choice displayed above. Their construction proofs are imported; the present task is their compatibility under covers.

Use and extent. This supplies the common primitives for S17 and S18 at every admitted scale. Equality is not asserted for independently chosen weakened systems. Allocate 4–5 proof pages to G and S16 together.

Source. 263-page master §10.9, §§15.2–15.3, Proposition 15.2, (10.45)–(10.46), (15.6)–(15.9). S3 Propositions 26.3(b),(c), 26.4 and 26.5, with the proofs in Chapter 26.

4. S17. Common lifted algebras and exact supported rows

Model-specific. Overview Target.

Objects and hypotheses. Take a top torus and its odd smaller quotient with sides $\ell_2 = J_c\ell_1$, and the related projected kernel systems. Require $\ell_1 > (31L+16)M$. Put $d_U = 2L+1$, $c_l = 2L$. For a finite connected covering union $\widetilde{X}$ of M-cubes, let $\operatorname{diam}(\widetilde{X})$ be the largest side of its smallest rectangular cube-box. Call it admissible when

$$\operatorname{diam}(\widetilde{X}) \le \ell_1 - 14d_U M, \qquad N(\widetilde{X}) = \operatorname{box}(\widetilde{X})^{\sim 6d_U}.$$

Here $X^{\sim a}$ means a successive neighboring M-cube enlargements. Images of such connected lifts are flat polymers. Ordinary, pair, unit and coordinate indices have their geometric supports and matching column measures on the two tori. The weakened kernels are restricted to the linked active components when parameters outside are zero; their rows read only that component's connection. The stronger H_k-row vanishing margin is retained.

Meaning of local agreement. For connected local families, agreement means equality after identifying the independent covering generators:

$$\iota_{\widetilde{X}} F_1(\operatorname{pr}_1 \widetilde{X}; A_1, C_1) = F_2(\operatorname{pr}_2 \widetilde{X}; A_2, C_2)$$

whenever the connection pairs have equal lifts on $N(\widetilde{X})$. The isomorphism is written from the first local algebra to the second. Local read dependence subsequently gives equality for pairs agreeing on the bonds actually read by the family.

Claim. Projection is injective on $N(\widetilde{X})$; any nonzero period separates its sites by at least $2d_U M + \eta$. Supported indices and incidences of total support diameter less than $2d_U M$ have unique common lifts. Pair separations and their six coordinate paths agree. Identifying independent covering generators defines a norm-preserving local Grassmann algebra isomorphism $\iota_{\widetilde{X}}$, preserving measures, degrees and bar indices. It includes chamber W-modes at zero, with no auxiliary Ψ there. Every small polymer $d_M \le L$ is flat; small polymers containing a fixed cube correspond across the two tori with identical volume and net length.

Let $\widetilde{Y}$ be admissible and $0 \le j \le 5d_U$. Parameters have equal lifts on cubes contained in $\widetilde{Y}$, and vanish outside their respective images. If the connections agree on $\widetilde{Y}^{\sim(j+d_U)}$, then every parameter-primitive row contained in $\widetilde{Y}^{\sim j}$ agrees exactly, has columns in $Y_i^{\sim(j+c_l)}$, and reads only bonds in $Y_i^{\sim(j+d_U)}$. For a connected $\widetilde{X} \subset \widetilde{Y}$, the local polymer kernels $M_k(X)$ for $k \ge 1$, and $S_{k,y}(X)$ at every level, also agree at common row/column lifts if the connections agree on bonds meeting $\widetilde{X}$.

The finite rows of the photon coordinate map, covariant pair quotient, nested averages, block resolvent $B_{k,y}$, interior and boundary reconstruction, mass form and first current agree as well. Their kernel equals the row coefficient divided by the common column measure. Adjustment, field scaling, weighted Wick integration and the normalized photon-coordinate marginal commute with ι.

Proof work. Use the explicit injective margin to rule out any distinct period representative in a local incidence. The common independent generators give the free-algebra isomorphism. For a supported row first restrict to its attached active component. Apply the signed image formula for the projected operator. All possible nonzero column lifts are in the same separated neighborhood, so only image $w = 0$ survives. A connected preimage of a flat polymer is one period translate of its lift. Row support forces this translate to be the chosen one; column support then forces the zero image in the signed kernel sum.

For finite rows check actual ranges and read bonds, including the reconstruction's boundary membership indicators. The normalized integration formulas are recorded below.

Finite formulas to retain. The photon coordinate rows have range at most $2L-1$ and do not read the connection. Pair evaluation has range at most M, reads its six paths, and has phase one within a cluster. Reconstruction retains the actual interior and boundary membership indicators. The

nested averages and finite block resolvent use their displayed short ranges, below one or at most $L-1$. At zero compare the whole four-neighbor F_0 stencil, not only its external edge; the identity kernel is 48δ. Mass and current rows retain the fine column measure $\eta^3/48$ and the first current's coefficient $1/96$.

The normalized common Wick moments are their ordering signs times $\nu^{-\#\text{pairs}}$, where ν is the common column measure; hence the factor is $48^{\#\text{pairs}}$ at zero. Unused modes integrate to one. The common photon marginal gives the same scalar box cost

$$\delta\epsilon_k^0 = -(2-2L^{-3})\log\left[(2\pi)^{-1/2}\int_{-p_{0,k}}^{p_{0,k}} e^{-z^2/2}\,dz\right],$$

where $p_{0,k} = (-\log e_k)^{p_0}$ is the prescribed cutoff. Its bound $0 \le \delta\epsilon_k^0 \le e_k^7$ is an imported step estimate.

Inputs. Common projected primitive system; component locality; polymer lifts; finite row formulas; weighted moments and scaled/adjusted local algebra. Exports: exact local row equality and the common free algebra. This is an equality statement, not an exponentially small finite-size error.

Source. 263-page master, pp. 163–168, §§15.4–15.6, Propositions 15.3–15.4, (15.10)–(15.20). S3 Propositions 27.1–27.4 and their proofs in Chapter 27, especially the supported-row and normalized-integration arguments. Allocate 4–5 proof pages.

5. S18. Equality of local nonlinear steps across odd covers

Model-specific. Overview Target.

Objects and hypotheses. Use one top system and its odd projected system on sides $\ell_2 = J_c\ell_1$, with both incoming and outgoing sides admissible and

$$\ell_1/L > (31L + 16)M.$$

A covering connected polymer is admissible if its box diameter is at most $\ell_1 - 14(2L + 1)M$; its images are flat. Compare their ordinary, pair and fluctuation generators using the norm-preserving common lifted algebra, with chamber measure $\eta^3/48$, and $1/48$ at the first step. The activities use exactly S09's class $\mathcal{A}_k^x$ on each torus, with its explicit analytic domains and generator transformation equations. The same real mass satisfies $|m| \le e_k^{3/4-8\varepsilon}$. The two input activities have norms at most $e_k^{1/4-7\varepsilon}$, vanish at $k = 0$, and agree on every flat lift when their two connection slots agree on its local read neighborhood. The exact inputs on both tori are the S15 maps with their stated density identity and bounds, S13's displayed finite contour definitions and domain inclusions, and S17's supported-row equalities. Use common $L, M, \alpha, \varepsilon, \kappa, e_k, \mu$ and fixed kernel constants. Their finite numerical prerequisites are explicitly listed in S05 (weakening/domain output), S08 (determinant gain), S09 (total connection), S11 (normalized scaling), S13 (finite localization) and S15 (scaled correction); the sole additional side requirement here is the one displayed above. The local construction consists of scalar extraction, shortest-chain quadratic adjustment, the two finite contour/parameter localizations, weighted fluctuation integration, the absolutely convergent connected union series, and scaling/reblocking.

The incoming scalar external families $\mathfrak{P}_k^{(1)}, \mathfrak{P}_k^{(2)}$ must also agree on flat lifts for the scalar-family conclusion below. They start at zero in T01.

Claim. The extracted scalar mass and vacuum contributions are equal, as is the photon box cost $\delta\varepsilon_k^0$. The local determinant primitive $E_k^d(X, B)$ is equal on each flat lift whenever B agrees on the bonds meeting that lift. The finite local fluctuation activities, their integrated Mayer activities and their local connected logarithms agree. Consequently the outgoing activity maps

$$E_i^+ = \mathcal{L}(RE_i)^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\mathrm{det},(i)} + E_k^{*,(i)}(m, E_i)$$

agree on every flat outgoing lift, where $\delta E_k^{\mathrm{det}}(X; A, C) = E_k^d(X, A + C) - E_k^d(X, C)$. The scalar recursion

$$\mathfrak{P}_{k+1}^{(i)}(Y, C') = \sum_{\overline{X}^L = LY} [\mathfrak{P}_k^{(i)}(X, C'_L) + E_k^{\mathrm{det},(i)}(X, C'_L)]$$

preserves this local agreement. All conclusions also hold without the external slot and commute with its restriction to zero. There is no error term or volume limit in this statement: the price is its explicit side margin and related primitive systems.

Proof work. Identify the extracted constant/quadratic coefficients and normalized integrals on each common small lift. For determinant traces all indices are supported in the flat polymer; the common finite B, Q, F_0 rows and local S kernels agree. Paired antiperiodic signs cancel. Here b_k is the finite block-resolvent parameter in the determinant formula. Use the integrable majorants $(b_k^2 + y^2)^{-1}$ at positive levels. At zero use the retained $(1 + y)^{-1}$ resolvent bound in the source's separate dy/y trace integrals on $[1, \infty)$ and dy integrals on $[0, 1]$, with the F_0S and $\gamma_5 S$ traces respectively.

Track the successive neighborhoods of the composed photon field: the reconstruction columns are in $U^{\sim 2d_U}$, coordinate-map columns in $U^{\sim(1+2d_U)}$, and root columns in $U^{\sim 3d_U}$. All lie in $N(U)$, so photon fields agree on the read neighborhood of every contour background. Then identify the fermion rows, first contour functions, their localization and reconstruction, the second functions and localization, in that order. Common coordinate marginals and weighted Wick moments identify the integrated Mayer activities. Termwise equality and absolute convergence identify their connected

logarithms: the even Mayer factors have the same incompatibility graph and hence the same Ursell coefficient. Convergence and its majorant are imported from I; local equality is proved here before summing.

For an outgoing lift $\widetilde{U}$, put $\widetilde{Y} = L\widetilde{U}$. Both reblocking sums have the same covering index set: connected $\widetilde{X} \subset \widetilde{Y}$ whose LM-closure is $\widetilde{Y}$. Compare the adjusted activity, determinant difference and local logarithm on each lift, then apply the common scaling. Locality extends the comparison from pullback connections to all pairs with equal local lifts.

Inputs. Exact local row/algebra comparison; determinant primitives; extraction/adjustment; finite localizations; connected convergence; exact nonlinear step. Exports: equality of local steps and counterterms for the later common-pair tuning argument. No equality on wrapping nonflat polymers is asserted. Uniform norm estimates alone do not prove this claim.

Source. 263-page master, pp. 168–172, §§15.7–15.9, Theorem 15.5, (15.21)–(15.32). S3 Propositions 27.5–27.6 and Chapter 27a: extraction, determinant, first localization and reconstruction, second localization, integrated Mayer activities and reblocking. Allocate 6–8 proof pages. Together with G, S16 and S17, the complete cover group occupies 14–18 pages.

6. T01.a. Stopping scales and availability of the steps

Model-specific. Overview Parent T01 Target.

Hypotheses and conclusion. Use the physical definitions and numerical conditions of T01. Write $R = hL^T$. For integer depth $s \geq 0$, set

$$\widehat{v}_s = v_K + \tfrac{s}{2}\log L, \qquad \widehat{r}_s = \lceil \widehat{v}_s^r \rceil_L, \qquad \widehat{n}_s = T + s - \log_L(M\widehat{r}_s),$$

where $\lceil\cdot\rceil_L$ is the least nonnegative integral power of L above its argument. Let $s_*(R,e)$ be the largest zero of $\widehat{n}_s$. Then

$$N_{\min}(R,e) = \max(0, s_* + 1 - \log_L h), \qquad k_*(R) = K - s_* \quad (N \geq N_{\min}(R,e)).$$

This is a sufficient threshold with a positive stopping level; it does not assert nonexistence of a zero-level stop below the threshold. The thresholds are nonincreasing and the stopping levels nondecreasing in R. The physical terminal quantities are independent of N. For $0 \leq j \leq k_*$,

$$\ell_j/(LM) \text{ is odd and at least } 13, \quad \ell_j \geq Mr_j, \quad r_j \geq r_* > 31L + 16,$$

$$0 < e_j \leq e_* \leq e_K \leq \bar{e}_1, \qquad 0 < \mu_j \leq \mu_* \leq \bar{m}h \leq 1.$$

For $j < k_*$, also $\ell_j \geq LMr_j$ and $\mu_j \leq L^{-1}$. At the stop,

$$\ell_* = Mr_*, \quad Mv_*^r \leq \ell_* < LMv_*^r, \quad \mu_*^{-1} < L\ell_*, \quad e_*^2\ell_* = e^2R.$$

At R_{top}, $s_* \in \{0,1\}$; at R^*,

$$s_* \in \{1,2\}, \quad a_* \in \{h/L, h/L^2\}, \quad r_* \in \{r_K, Lr_K\}, \quad L^{-3} < \mu_* \leq L^{-1}.$$

Arithmetic proof. The slow-growth condition in (1) implies $\widehat{r}_{s+1}/\widehat{r}_s \in \{1, L\}$, and, for $s = 1,2,3$, gives $\widehat{r}_s \in \{r_K, Lr_K\}$. Thus $\widehat{n}_s$ has increments zero or one, starts at a nonpositive integer and tends to infinity. Its last zero is finite. Reading depth in the reverse order shows that $K - s_*$ is the first stopping level once $K \geq s_* + 1$. Changing R by a factor L shifts $\widehat{n}_s$ by one; the same sequence comparison gives monotonicity. The stop fixes $a_* = hL^{-s_*}$, hence all terminal physical quantities. On R^*, the values of $\widehat{n}_s$ at $s = 0,1,2,3$ are -1, an element of $\{-1,0\}$, an element of $\{0,1\}$, and an element of $\{1,2\}$. The zero-or-one increment rule gives last zero 1 or 2. The top sequence is larger by one and has last zero 0 or 1.

Applicability. The actual step contracts in I are standing hypotheses here. The arithmetic verifies their remaining scale and side conditions: both ℓ_j and ℓ_{j+1} are admitted, the incoming mass is at most L^{-1}, and the next charge is below its ceiling. The primitive families at each scale are the top families of S16 and their odd projections. For a compared step on a smaller torus,

$$\ell_{j+1}(R) \geq Mr_{j+1} > (31L + 16)M.$$

This is exactly the outgoing-side margin required by S18. It is the reason comparison can be used through the smaller stop.

Source. 263-page master §16.1, Proposition 16.1 and (16.1)–(16.7). S3 §G4.0.2 retains S1 Proposition 19.1 with $h' = h$, old mass depth zero and upper running mass one; the supplied arithmetic proof is S1 Chapter 19. S3 Proposition 26.6 and Chapter 28, §0.3 verify step applicability. Included in T01's 7–10 proof pages.

7. T01.b. Continuation and the one-torus terminal-zero pair

Model-specific. Overview Parent T01 Target.

Inputs and one-torus result. Fix an eligible torus and cutoff as in T01.a, with the actual maps and normalized bounds in T01. There is a unique controlled real bare pair for which both scalar couplings vanish at its stop. Its activity, mass and vacuum obey (5). The pair used in the final theorem is this pair on R_{top} alone. The construction has the following steps.

Continuation. Suppose the already computed masses remain in $|m_j| \le w_j^m$. Set $A_j = \|\mathscr{E}_j\|_j^x / w_j^E$. The size estimate, including its external slot, gives

$$A_{j+1} \le A_j/16 + 2e_{j+1}^{\varepsilon}, \qquad A_0 = 0.$$

The supplied geometric sum yields

$$A_j \le (32/15)e_j^{\varepsilon} \le 4e_j^{\varepsilon} \le 1.$$

Consequently every next step is defined in its radius-four ball. Restriction gives $E_j = \mathscr{E}_j|_{C=0}$, and the real class and scalar recursion are preserved. The initial step uses the chamber conventions of I, with incoming activity fixed to zero.

The finite-sequence contraction. Use the closed unit set, with $E_0 = 0$, in the complete real product space with norm

$$\|(m,E)\| = \max_{0\le j\le k_*} \max\{|m_j|/w_j^m,\ \|E_j\|_j/w_j^E\}.$$

Define the backwards mass and forwards activity map by

$$\widehat{m}_j = L^{-1}m_{j+1} - m(E_j)\ (j<k_*), \qquad \widehat{m}_{k_*} = 0, \qquad \widehat{E}_0 = 0, \quad \widehat{E}_{j+1} = \Phi_j(m_j, E_j).$$

At zero the activity coordinate has only its prescribed value zero. The charge hypothesis gives

$$\sum_{j=0}^{k_*} (c_{\mathrm{fl}} + 2)e_j^{\varepsilon} \le 1/16.$$

The normalized mass component maps the unit set into radius 5/16; the activity component maps it into radius 1/8. Their difference estimates have the same bounds, so the whole map contracts by 5/16. The deterministic local term cancels in the difference since it is independent of the sequence. The contraction theorem gives the unique fixed point in this set. Continuation identifies its no-slot variables with the actual trajectory and supplies the full external-slot activities.

Vacuum and sharper bounds. Recover the unique vacuum with terminal value zero by

$$\epsilon_j^v = -\sum_{i=j}^{k_*-1} L^{-3(i-j)}[\epsilon_v(E_i) + \delta\epsilon_i^0].$$

This solves the exact vacuum recursion. Every controlled terminal-zero trajectory lies in the product unit set by continuation, so fixed-point uniqueness and this scalar formula prove uniqueness among controlled pairs.

Write $B_j = |m_j|/w_j^m$. The backwards mass recursion now gives

$$B_j \le B_{j+1}/4 + 1/16, \qquad B_{k_*} = 0, \qquad B_j \le 1/12 \le 1/8.$$

In the vacuum sum, extraction contributes at most $(64/15)c_{\mathrm{fl}}e_j^{1/4-6\varepsilon}$, and the box costs at most $e_j^{1/4-6\varepsilon}/48$. Their sum is bounded by $5c_{\mathrm{fl}}e_j^{1/4-6\varepsilon}$. Continuation has already supplied the asserted activity bound.

Source. 263-page master §§16.2–16.3, Proposition 16.2 and Theorem 16.3. S3 Propositions 28.1–28.2, with the continuation, finite-sequence contraction and vacuum sum in Chapter 28. Together with T01.a, allocate 7–10 proof pages.

Appendix A. Detailed proof tasks and supporting statements

The following cards state the imported contracts and recall the shared activity definitions. They introduce no additional conclusion beyond the opening hypotheses.

I. Imported primitive and nonlinear-step results

Model-specific. Overview Target.

Role in the target. The opening theorem assumes these results with their full domains and numerical prerequisites. The links resolve to QEO1's contracts for the actual maps.

Primitive construction. S03 and S04 give the photon block minimizers, coordinate maps, covariance and positive resolvent root. Their focused treatment is MBO1. S05 gives the weakened-system properties and domains; S06 gives the chamber inverses and weighted walks, including the resolvent parameter. The fermion treatment is FBO1. These imports include all mass, block-size and charge restrictions. In S16 they are imposed at decay $\kappa_w + \kappa_0$ before projection, with the required analytic domains, active components and row-vanishing conditions.

Activity class and linear operations. S09 gives the class in D, its complete real spaces, total-connection inclusion, real linear scalar extraction and shortest-chain quadratic adjustment. S10 gives the actual interior/boundary reconstruction rows and bounded substitution. S11 gives normalized reblocking, with all scaled-domain inclusions. These are treated in RNO1. For the flow argument one uses the relaxed consequence

$$\|\mathcal{L}(RE)^{\mathrm{nat}}\|_{j+1} \le c_{\mathrm{fl}} L^{-1/4+2\varepsilon} \|E\|_j .$$

In particular its normalized factor is $c_{\mathrm{fl}} L^{-3/8+11\varepsilon/2} \le 1/16$. Restriction to zero external connection commutes with each operation.

The nonlinear maps. S08 supplies the local determinant primitive and its scaled gain. Its no-slot source term satisfies

$$\|\mathcal{L}E_j^{\mathrm{det}}\|_{j+1} \le e_{j+1}^{1/4-2\varepsilon}.$$

S12 gives the exact density identity. S13 supplies the two finite contour/parameter localizations with their domain inclusions. S14 gives the absolutely convergent connected union series and its normalization. S15 combines them into the actual step and supplies its reality and estimates. On the radius-four data ball the correction has size $\|E_j^*(m,E)\|_{j+1} \le e_j^{1/4-6\varepsilon}$. On the unit ball it obeys

$$\|E_j^*(m,E) - E_j^*(\widetilde{m},\widetilde{E})\|_{j+1} \le e_j^{1/4-6\varepsilon} \left(\frac{|m-\widetilde{m}|}{w_j^m} + \frac{\|E-\widetilde{E}\|_j}{w_j^E} \right).$$

The same source-size bounds hold with the external slot, and restriction gives the no-slot map. The exact scalar recursions are (2), with $0 \le \delta\epsilon_j^0 \le e_j^7$. Their normalized consequences are recorded in the opening statement.

First step. At zero there are $192\ell_0^3$ incoming chamber pairs with weight $1/48$, and $4\ell_1^3$ outgoing pairs. The incoming activity is fixed to zero, the zero-level chamber stencil is used in full, and its identity covariance is 48δ. These definitions are part of the exact first-step formulas, not a limiting interpretation of a positive-level step.

Parameters. For each linked contract, retain its displayed finite inequalities: the weakened-domain conditions in S05, determinant gain in S08, total-connection bounds in S09, reconstruction weights in S10, normalized scaling in S11, finite localization in S13, cluster smallness in S14, and scaled correction in S15. They are standing assumptions of T01; its charge inequalities alone are not a replacement for them. Their simultaneous selection is the ordered-constants verification in 263-page master §27, Proposition `ordered-constants`, outside this focused outline.

Source. 263-page master §§10–14 and §16.2; the original S3 construction and Chapter 28, §0.

D. The activity space and its field domains

SETUP. Overview Target.

Scope. The cover comparison identifies these independent generators and their weighted norms. The tuning argument uses the complete real activity spaces defined by them. All fields in this card are at one fixed level; its mesh is denoted by $\eta = L^{-k}$, as in the primitive construction. It is distinct from the physical spacing L^{-N} in T01.

Chamber labels and evaluation. A chamber site is $x = (c, g)$, where $c \in \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with $\eta = L^{-k}$ for a nonnegative integer k, and g ranges over the 48 signed permutation matrices in dimension three. There are four spin components and separate barred and unbarred generators. The three internal neighbours replace g by gs_i, where s_0 flips the third coordinate, s_1 exchanges the last two, and s_2 exchanges the first two. The external neighbour is $x^{(3)} = (c + \eta g e_1, g r_{e_1})$, where r_{e_1} flips the first coordinate. This neighbour convention fixes the external pairs used below. For a connection B, $U_B(x, x')$ is the exponential of ie_k times the arithmetic mean of its six coordinate-order path sums, each with the mesh factor η. It is one within a photon cluster. The pair generators remain independent until the evaluation

$$\chi(x, x') = \frac{U_{A+C}(x, x')\psi(x') - \psi(x)}{\max(|c - c'|_\infty, \eta)^\alpha}$$

and its barred version with inverse transport. Statements about an evaluated activity refer to this substitution.

The common setting. Define the activity space $\mathcal{A}_k^{x,\mathbb{C}}$, with real subspace $\mathcal{A}_k^x$. Fix odd $L \geq 5$, $\eta = L^{-k}$, $M = L^m \geq L^2$, $\ell/(LM)$ odd and at least 13, $0 < \varepsilon \leq 1/200$, $0 < \alpha < 1$, and $0 < e_k < 1$. An M-polymer is small if $d_M(X) \leq L$, with an injective covering lift. Put $\mathrm{Vol}(X) = M^3|X|_M$. Balanced means that each monomial has equally many barred and unbarred generators. An activity $E(X; A, C)$ is a balanced polynomial in independent ordinary and pair Grassmann legs, with holomorphic coefficients reading only bonds meeting X. Ordinary legs are over X; an allowed pair has all six coordinate-order paths in X, or is an external nearest-neighbor pair with an endpoint over X. Its separation is $d_X = \max(|c - c'|_\infty, \eta)$, and the pair range is $0 < d_X \leq M$; intracluster pairs are included. Index measures are $\eta^3/48$ and $(\eta^3/48)^2$, while generator weights are $h_1 = e_k^{-1/4}$, $h_2 = e_k^{-1/4+\varepsilon}$. Ordinary kernels use the antiperiodic convention in each chamber argument; pair indices are simultaneous period classes of their two endpoints. The coefficient norm of the separately antisymmetric kernels is

$$\|E(X; A, C)\|_{h_k} = \sum_{n,m} \frac{h_1^n h_2^m}{n!m!} \int |E_{nm}(X; A, C; \xi; \zeta)| \, \mathrm{d}\xi \, \mathrm{d}\zeta.$$

To specify its domains, put

$$\begin{aligned} \partial_\nu B_\mu(c) &= \eta^{-1}[B_\mu(c + \eta e_\nu) - B_\mu(c)], \\ \|B\|_1 &= \sup |B_\mu(c)|, \quad \|B\|_2 = \sup |\partial_\nu B_\mu(c)|, \\ \|B\|_3 &= \sup_{\mu,\nu} \sup_{0<d(c,c')\leq 1} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c, c')^\alpha}, \qquad \theta_i = e_k^{-3/4+i\varepsilon}. \end{aligned}$$

The distance is periodic sup distance; a local norm uses only its contained bonds and difference cells. $R_k = \{B : \|B\|_i < \theta_i,\ i = 1, 2, 3\}$. A field belongs to $\widetilde{R}_k$ if it is $A_0 + A_1$, A_0 real, $A_1 \in R_k$, and on every enlarged cube of side $(2L + 1)M$, $A_0 = A_\square + d\lambda_\square$ with real $A_\square, \lambda_\square$ and $\|A_\square\|_{i,\square^\dagger} < \theta_i$. The external domain is

$$\mathfrak{X}_k = \{C : \|C\|_1 < e_k^{-1+2\varepsilon},\ \|C\|_2 < e_k^{-1+3\varepsilon}\}.$$

The activity norm is precisely

$$\|\mathscr{E}\|_k^x = \sup_X \exp(\kappa d_M(X)) \sup_{(A,C)\in \widetilde{R}_k\times\mathfrak{X}_k} \|\mathscr{E}(X;A,C)\|_{h_k}.$$

Here $\mathscr{E}$ denotes the full activity previously written $E(X;A,C)$. For its restriction $E=\mathscr{E}|_{C=0}$, define $\|E\|_k$ by the same supremum with $C=0$. Hence $\|E\|_k \le \|\mathscr{E}\|_k^x$. The no-slot real space is defined by the corresponding symmetry equations. At level zero the flow's activity coordinate is fixed to zero; the chamber fluctuation variables still occur in its actual first-step integral. Assume the polymer counting bound $\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \le K_0$, $\kappa \ge \kappa_0$.

Here is the complete symmetry package. Use $\gamma_\mu = \sigma_3\otimes\sigma_\mu$, $\gamma_5 = \sigma_1\otimes 1_2$, $\mathsf{C} = i\gamma_2$, and $\varkappa = \gamma_5\mathsf{C}$. For a spin lift S generated by the four γ-matrices and a translation $a\in M\mathbb{Z}^3$, let $\widetilde{g}(c,g) = (r(S)c+a, r(S)g)$, where $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Define

$$a_{\widetilde{g}}\psi(x) = S\psi(\widetilde{g}^{-1}x), \qquad a_{\widetilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\widetilde{g}^{-1}x).$$

For pair generators transform both endpoints and use the same spin matrices. G_λ multiplies an unbarred ordinary or pair leg by $\exp(ie_k\lambda(c))$, with c its first photon point, and a barred leg by its inverse. Define on ordinary and pair legs

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \quad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \quad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

with $\mathbf{K}$ antilinear. The complex activity class requires

$$\begin{aligned} a_{\widetilde{g}}E(\widetilde{g}X;\widetilde{g}A,\widetilde{g}C) &= E(X;A,C),\\ \mathrm{Cg}\,E(X;-A,-C) &= E(X;A,C),\\ G_{\lambda+\lambda'}E(X;A^1,C^1) &= E(X;A,C) \end{aligned}$$

whenever the two pairs are in the domain and $A^1 = A - d\lambda$, $C^1 = C - d\lambda'$ on bonds at distance at most one from X, for real λ,λ'. Its real subspace additionally requires

$$\mathbf{K}E(X;-\overline{A},-\overline{C}) = E(X;A,C).$$

All equations act on independent generators before evaluation. For the reconstruction application in QEO1/S10, the explicit conditions

$$e_k^{\varepsilon} \le \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^{\varepsilon} \le 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \le 1$$

give $\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k$, where $\widetilde{G}_k$ has a global complex bound $e_k^{-1+\varepsilon}$ and local real-gauge sup bounds $e_k^{-1+\varepsilon}$ on every $11M$ cube. They also give $e_k\|\operatorname{Im}(A+C)\|_\infty \le 1$ and $e_k\|d(A+C)\|_\infty \le 1$.

Source. 263-page master §12.1; QEO1 S09. This is shared setup, not a separate estimate.